

SAINT ANDREW'S JUNIOR COLLEGE

PRELIMINARY EXAMINATION PAPER 2

MATHEMATICS

Higher 2

9740

Tuesday

9 September 2008

3 hours

Additional materials : Answer paper
List of Formulae(MF15)
Cover Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Answer **all** the questions. Total marks : 100

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically state otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematic steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of 5 printed pages including this page.

[Turn over

Section A: Pure Mathematics [40 marks]

1. Relative to a fixed origin O , the vector equations of the two lines l_1 and l_2 are

$$l_1: \mathbf{r} = 9\mathbf{i} + \alpha\mathbf{j} + 4\mathbf{k} + t(-8\mathbf{i} - 3\mathbf{j} + 5\mathbf{k})$$

and

$$l_2: \mathbf{r} = -16\mathbf{i} - 3\mathbf{j} + 10\mathbf{k} + s(\mathbf{i} - 4\mathbf{j} + 9\mathbf{k})$$

where α is a constant.

The two lines intersect at the point A .

- (i) Find the value of α . [3]
 (ii) Find the position vector of A . [1]
 (iii) Prove that the acute angle between l_1 and l_2 is 60° . [2]

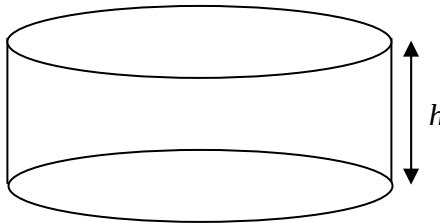
The line l_1 passes through the point $B(1, -1, 9)$ and the line l_2 passes through the point $C(-17, 1, 1)$.

- (iv) Find the exact area of triangle ABC . [4]

2. The complex numbers z_1 and z_2 are given by $z_1 = 1 + \sqrt{3}i$ and $z_2 = iz_1$.

- (i) Find the exact value of the modulus and argument of z_2 . [1]
 (ii) Label the points representing z_1 and z_2 on an Argand diagram. [1]
 On the same Argand diagram, sketch the following loci.
 (a) $|z - z_2| = |z_1 - z_2|$ [2]
 (b) $\arg(iw - z_2) = \arg z_1$ [3]
 Find the least possible value of $|z - 2| + |w - 2|$ [2]

3. A fish tank with an oval base and vertical height h is initially full of water. Water leaks out of a small hole in the base of the tank at a rate, which, at any time t , is proportional to the depth of the water x at that instant.



- (i) Show that the differential equation, in terms of x and t , can be written as $\frac{dx}{dt} = -kx$ where k is a positive constant. [2]
 (ii) If the tank is exactly half empty in 2 hours, find the depth of water in an hour, giving the exact answer in terms of h . [8]

[Turn over]

4. The functions f and g are defined by

$$f : x \mapsto \frac{1+x}{1-x}, -1 \leq x < 1$$

$$g : x \mapsto \frac{1}{1+x^2}, x > -1$$

- (i) State with reasons whether the composite functions fg and gf exist. If it exists, find its rule, domain and range. [6]
- (ii) Explain why the inverse function g^{-1} does not exist. State the maximal domain of g in the form of an interval $[a, \infty)$, where a is a constant to be determined, so that g^{-1} exists. For this restricted domain of g , find g^{-1} in a similar form, stating clearly its domain and range. [5]

Section B: Statistics [60 marks]

5. A group of 8 friends sit together in a circle. Alice, Betty and Candy are three members of the group of friends.
- (i) If Alice refuses to sit beside Betty unless Candy sits on the other side of Alice as well, how many possible seating arrangements can there be? [3]
- (ii) 2 latecomers then come to join the group, and they have to sit apart from each other. How many possible seating arrangements can there be, bearing in mind the condition from (i)? [3]
6. Each of a random sample of 10 students are asked about the average number of minutes spent on doing mathematics tutorials in a week (x), and their percentage score for the mathematics final examination (y). The results are tabulated below:

x	20	35	45	60	70	80	100	110	120	140
y	16	25	35	50	60	65	70	75	80	85

- (i) Find the equation of the regression line of y on x . [1]
- (ii) Find the linear product moment correlation coefficient between y and x , and comment on the relationship between x and y . [2]
- (iii) Making use of your answer from part (i), find the equation of the regression line of x on y . [2]
- (iv) Use the appropriate regression line to estimate the percentage score of a student who spends 10 minutes doing mathematics tutorial in a week. Comment on the reliability of the estimate. [2]

7. With the upcoming movie “Dragonball” in 2009 by 20th Century Fox, a toy company is beginning to manufacture a Design A soft toy and Design B soft toy based on the main character. It was discovered that an average of 5 defective Design A soft toy and an average of 3 defective Design B soft toy are manufactured in a day.
- (i) Find the probability that there are at most altogether 12 defective toys manufactured in a day. [3]
 - (ii) Using a suitable approximation, find the probability that there are exactly 350 days out of 365 days in a year where there are at most 12 defective toys manufactured in a day. [4]
8. A student has requested for a testimonial to apply for a job. She estimates that there is an 80% chance of getting the job if she receives an excellent testimonial, a 40% chance if she receives a moderately good testimonial and a 10% chance if she receives a fair testimonial. She further estimates that the probabilities that the testimonial will be excellent, moderate or fair are 0.7, 0.2 and 0.1 respectively.
- (i) Draw a tree diagram to illustrate the information and calculate the exact probability of the student getting the job. [2]
 - (ii) Given that she receives the job offer, what is the exact probability that she received an excellent testimonial? [2]
 - (iii) Given that she did not receive a moderate testimonial, what is the exact probability that she did not receive a job offer? [2]
 - (iv) If there are n students, who have the exact same chance of obtaining the kind of testimonials as this student, what is the probability of at least 1 student obtaining an excellent testimonial? Leave your answer in terms of n . [2]
9. On the first day of an IT fair, a sales representative sets up a booth to sell the latest model of Nintendo Wii. Assuming that the time taken to sell the product follows a normal distribution with mean μ minutes and variance 5 minutes.
- (i) Given that the probability of taking less than 20 minutes to sell a Wii set is 0.55, find μ . [4]
 - (ii) Find the probability that the time to sell 10 Wii sets is less than 3 hrs. [2]
 - (iii) On the second day, the sales representative decided to bring in another product w that is twice as saleble as a Wii set. Find the probability that the total time taken to sell one set of each product is not more than 30 minutes. [4]

- 10.** An employer has commissioned an opinion polling organisation to undertake a survey of staff responses to the proposed changes in the pension scheme. The staff are categorised as management, professional and administrative, and it is thought that there might be considerable differences of opinion among the categories. There are 60, 140 and 300 staff respectively in the categories. The budget for the survey allows for a sample of 40 members of staff to be selected for in-depth interviews.

- (i) State one disadvantage of selecting a simple random sample from all the staff. [1]
- (ii) Describe an alternative sampling method which would be better in this case. [3]

The opinion polling organisation needs to estimate the average salary of staff. They took a sample of 50 people and found out that $\sum (x - 3000) = -1360$, $\sum (x - 3000)^2 = 8,687,200$.

- (iii) Find the unbiased estimates of the population mean and variance. Leave your answers to 2 decimal places. [3]
- (iv) Find the probability that the mean salary of 60 people taken at random is more than \$3,000. [3]

- 11.** A stretch of the SJE Expressway near a town centre has a speed limit of 70 km/h. Before the introduction of speed cameras, the vehicles using this Expressway had a mean speed of 81 km/h.

Following the introduction of speed cameras on this Expressway, ten drivers were fined for exceeding the speed limit. Their recorded speeds, in km/h, were:

93.7, 79.9, 86.0, 102.8, 84.9, 95.3, 89.6, 88.2, 93.0, 106.9

- (a) John, a representative of a motoring organization, stated that he has examined the data and found significant evidence that the mean speed of vehicles has increased since the introduction of the speed cameras. He therefore claims that the road would be much safer if the cameras were removed.
- (i) Assuming that the recorded speeds may be regarded as a random sample from a normal distribution, show that there is evidence at the 1% level of significance, that the mean speed exceeds 81 km/h. [4]
- (ii) Explain why John's claim is not valid. [2]
- (b) (i) To investigate whether the mean speed of vehicles has increased since the introduction of speed cameras, the speeds of a random sample of 120 vehicles using the expressway are recorded and are found to have a mean of 82.1 km/h and a standard deviation of 6.9 km/h. Using this second sample, and a 5% significance level, examine whether there is evidence that the mean speed of vehicles now exceeds 81 km/h. [5]
- (ii) State giving a reason, whether it is necessary to assume a normal distribution for the test to be valid. [1]

End of Paper

[Turn over